

Corpus Linguistics for Korean with Universal Dependencies: Concepts and Hands-On Practice

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Outline of Workshop

- Part 1
 - Corpus linguistics
 - Universal Dependencies
 - Practice & discussion
- Part 2
 - Annotation
 - Frequency extraction
 - Concordance line generation

Quick questions

- *What is your experience with corpora (e.g., **모두의 말뭉치**) and corpus tools (e.g. AntConc, Ngram viewers)?*
- *How familiar are you with Korean-specific POS tag sets (e.g. Sejong: NNG, JKS, XSV) and dependency relations?*

Corpus Linguistics: Broad Foundations

- Core pillars:
 - **Empirical** : studying language through large, principled collections of naturally occurring spoken and written text
 - **Usage-based**: moving beyond binary grammaticality judgements to examine distributional properties, co-occurrences, register/genre variations, and probabilistic patterns
 - **Descriptive**: observing what speakers *actually* produce in real communicative contexts rather than what normative grammars state they *should* produce

Counting Units: Token, Type, Lemma

- **Tokens** (running words), **types** (unique forms), **lemmas** (canonical dictionary base forms)
- Word decomposition by eojeol (a whitespace-based unit) or morpheme

Ku mal-i kwungkumha-acy-ess-eyo.
 That remark-NOM curious-INCH-PST-POL
 “[I] became curious about that remark.”

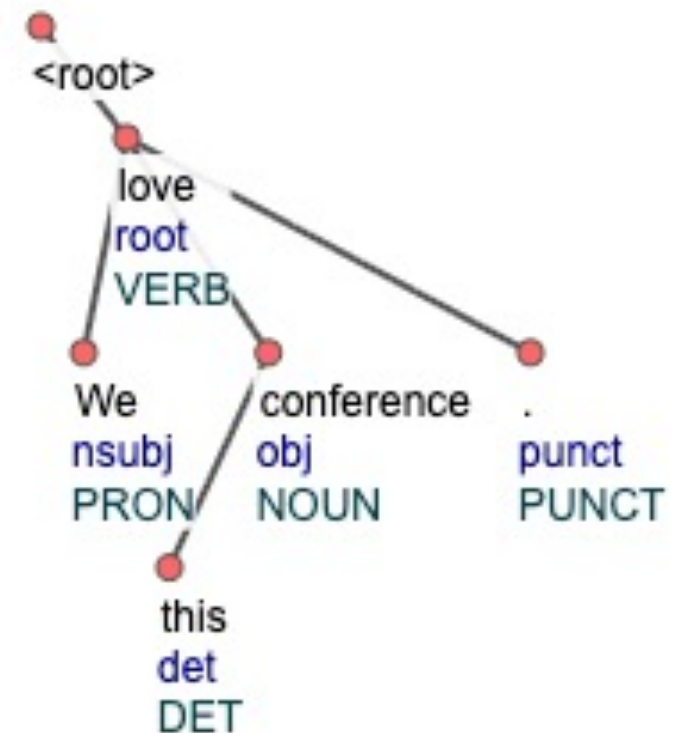
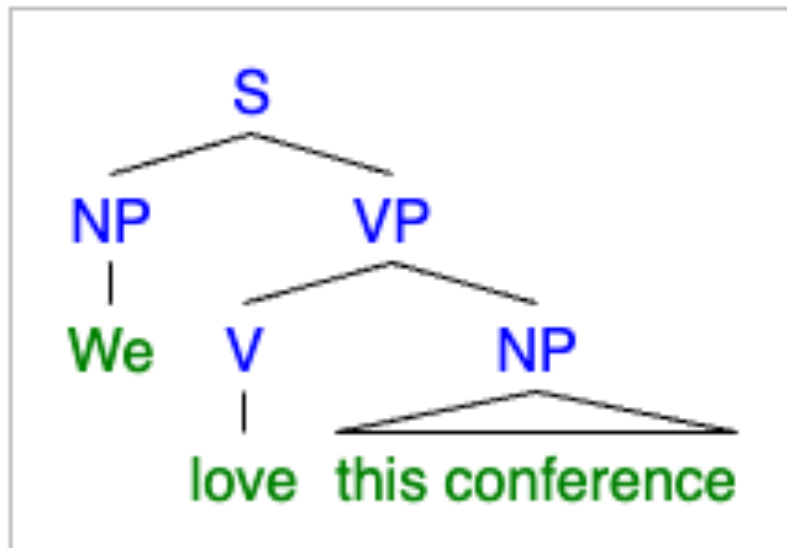
Morpheme	Function	Description
<i>ku</i>	Demonstrative	“that”
<i>mal</i>	Noun	“remark, word”
<i>-i</i>	Nominative case particle	subject marker
<i>kwungkumha</i>	Adjective stem	“to be curious”
<i>-acy-</i>	Derivational morpheme	Indicates change of state (“to become”)
<i>-ess-</i>	Past tense marker	Past tense
<i>-eyo</i>	Predicate ending, polite	Polite ending

Counting Units: Token, Type, Lemma

- **Tokens** (running words), **types** (unique forms), **lemmas** (canonical dictionary base forms)
- Word decomposition by *eojeol* (a whitespace-based unit) or morpheme
 - An *eojeol*-level approach reflects broader lexicogrammatical properties
 - A morpheme-level approach captures how content and functional morphemes interact

Universal Dependencies

Constituency grammar (phrase structure) vs. dependency grammar (head-dependent arcs)



What is UD? Overarching question

- How do we obtain language data...
 - which are good quality;
 - which capture both language-universality and language-specificity;
 - which are informative; and
 - which are useful for various purposes?

What is UD?

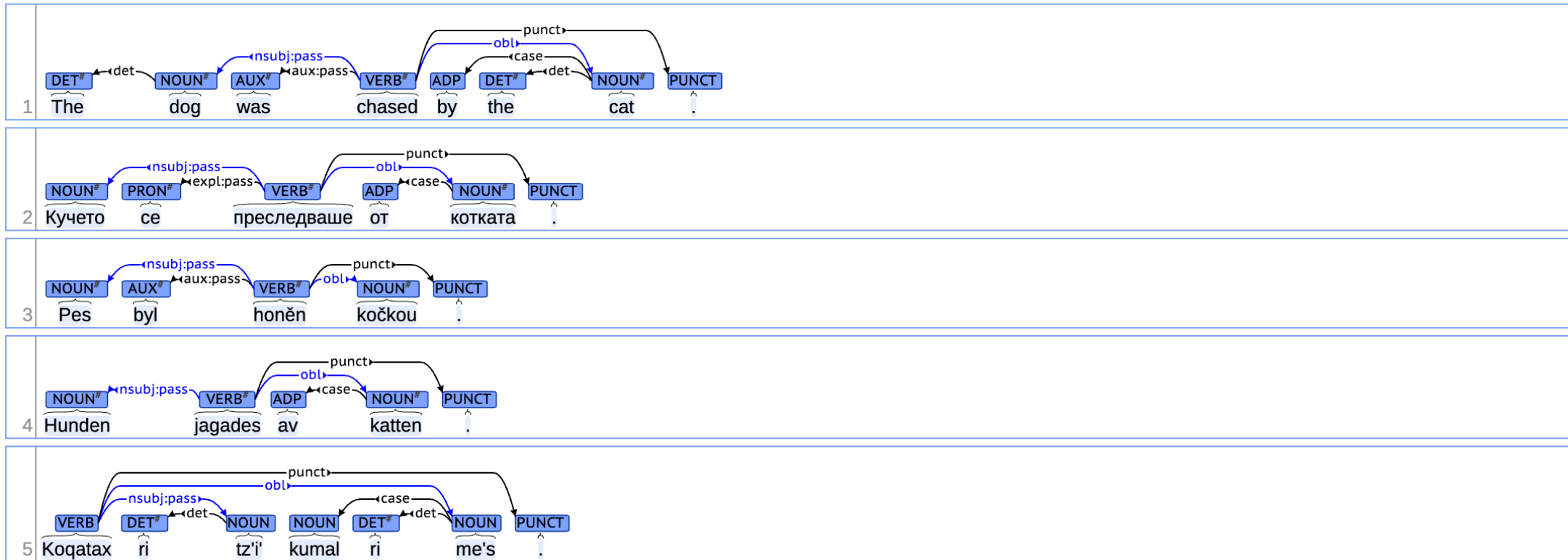
- A framework for consistent annotation of grammar (parts of speech, morphological features, and syntactic dependencies) across different human languages
 - Goal: facilitating multilingual parser development, cross-lingual learning, and parsing research from a language typology perspective
 - General philosophy: to provide a universal inventory of categories and guidelines to facilitate consistent annotation of similar constructions across languages, while allowing language-specific extensions when necessary

What is UD?

- Design principles
 - UD needs to be satisfactory on linguistic analysis grounds for individual languages
 - UD needs to be good for linguistic typology: It should bring out crosslinguistic parallelism across languages and language families
 - UD must be suitable for rapid, consistent annotation by a human annotator
 - UD must be easily comprehended and used by non-linguist users
 - UD must be suitable for computer parsing with high accuracy
 - UD must support well downstream language understanding tasks

What is UD?

The general philosophy of UD is to propose a universal inventory of categories and guidelines to facilitate consistent annotation of similar constructions across languages, while allowing for typological differences and using language-specific extensions when necessary. This is illustrated in the following parallel examples from English, Bulgarian, Czech, Swedish, and K'iche'. The main grammatical relations shown in blue (involving a passive verb, a nominal subject and an oblique agent) are the same, but the concrete grammatical realization varies.



de Marneffe et al. (2021)

- Basic tenets
 - Linguistic representation and information packaging
 - Three fundamental phrasal units
 - **Nominal**: the primary means for referring to entities
 - **Clause**: the primary means for referring to events
 - **Modifier**: the canonical attributive modifiers of nominals, clauses, and other modifiers

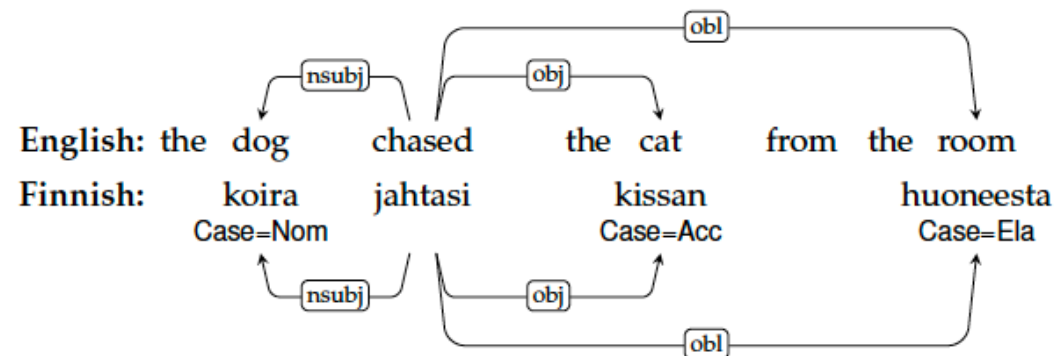


Figure 2
Simplified UD annotation for equivalent sentences from English (top) and Finnish (bottom).

de Marneffe et al. (2021)

- Basic tenets
 - Hierarchical structure via dependency grammar
 - **Phrase** = **head** (lexical or content word) + **dependents**
 - **Dependency**: a binary asymmetrical relation, which is represented through diagrams by an arrow from the head to the dependent
 - Main predicate of a clause = root
 - UD tree represents a sentence's observed surface predicate–argument structure rather than necessarily capturing phrase-internal syntactic constituency

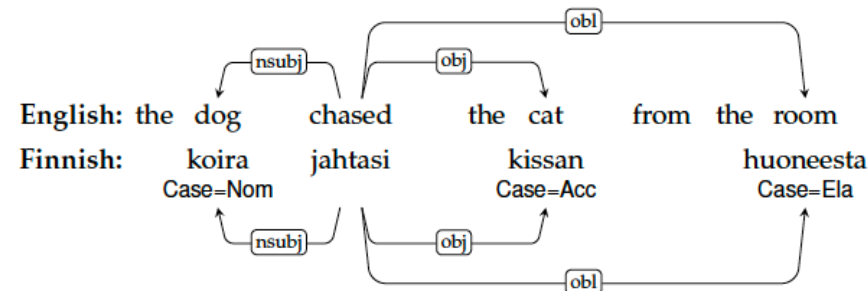


Figure 2
Simplified UD annotation for equivalent sentences from English (top) and Finnish (bottom).

de Marneffe et al. (2021)

- Basic tenets
 - **Words** as basic units
 - The morphosyntactic notion of word does not always coincide with orthographical or phonological units e.g., clitics, compound words
 - Words that enter into the main syntactic relations are autosemantic: content words with an independent meaning
 - Typically verbs, nouns, adjectives, or adverbs, as well as corresponding pro-forms with a contextually determined referential meaning
 - The distinctions between content words and function words, and between function words, clitics, and inflectional morphemes, are not always clear-cut
 - Grammaticalisation (e.g., “let us” vs. “let’s”)

de Marneffe et al. (2021)

- Key components
 - **Part-of-speech categories**
 - Each word is assigned a syntactic category based on its role in the sentence
 - Morphological features
 - Provide detailed information on word forms and inflections, which are not captured by POS tags and dependency relations
 - Dependency relations
 - Each word is assigned a grammatical function based on its role in the sentence

Table 1

Universal part-of-speech tags (UPOS). Typos and abbreviations are given the category of the unabbreviated or correct word.

Traditional POS	UPOS	Category
noun	NOUN	common noun
	PROPN	proper noun
verb	VERB	main verb
	AUX	auxiliary verb or other tense, aspect, or mood particle
adjective	ADJ	adjective
	DET	determiner (including article)
	NUM	numeral (cardinal)
adverb	ADV	adverb
pronoun	PRON	pronoun
preposition	ADP	adposition (preposition/postposition)
conjunction	CCONJ	coordinating conjunction
	SCONJ	subordinating conjunction
interjection	INTJ	interjection
–	PART	particle (special single word markers in some languages)
–	X	other (e.g., words in foreign language expressions)
–	SYM	non-punctuation symbol (e.g., a hash (#) or emoji)
–	PUNCT	punctuation

de Marneffe et al. (2021)

- Key components

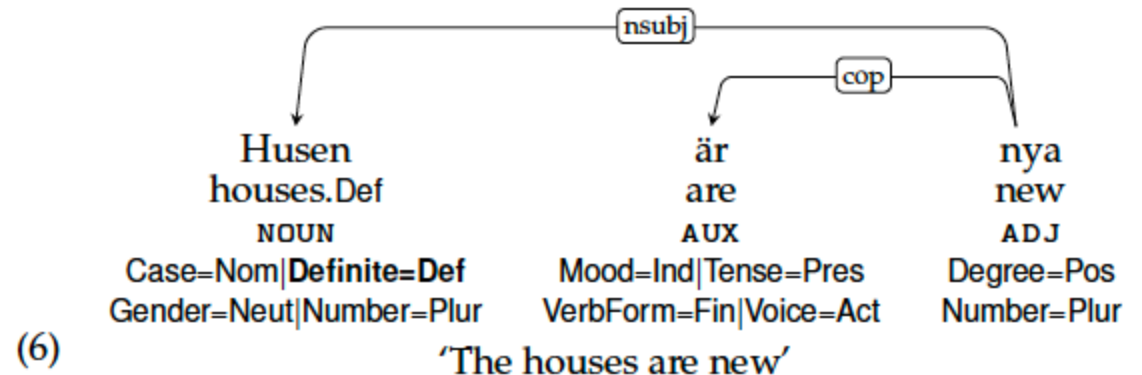
- Part-of-speech categories
 - Each word is assigned a syntactic category based on its role in the sentence

- **Morphological features**

- Provide detailed information on word forms and inflections, which are not captured by POS tags and dependency relations

- Dependency relations

- Each word is assigned a grammatical function based on its role in the sentence



de Marneffe et al. (2021)

- Key components

- Part-of-speech categories
 - Each word is assigned a syntactic category based on its role in the sentence
- Morphological features
 - Provide detailed information on word forms and inflections, which are not captured by POS tags and dependency relations
- **Dependency relations**
 - Each word is assigned a grammatical function based on its role in the sentence

Table 4
Typology of the syntactic relations.

Head \ Dependent	Nominals	Clauses	Modifier words	Function words
Clausal core arguments	nsubj obj iobj	csubj ccomp xcomp		
Clausal non-core arguments	obl vocative expl dislocated	advcl	advmod discourse	aux cop mark
Nominal dependents	nmod appos nummod	acl	amod	det clf case
Coordination	MWE	Loose	Special	Other
conj cc	fixed flat compound	list parataxis	orphan goeswith reparandum	punct root dep

de Marneffe et al. (2021)

- Key components
 - More about dependency relations
 - Core arguments of a predicate vs. oblique modifiers (all other dependents at the clause level)
 - UD assumes that all languages have a way of identifying usually two core arguments and reserves the relations of **subject** and **object** for these arguments
 - While status as a core argument is decoupled from the semantic role of a participant, there is a correlation
 - *Agent and patient/theme roles of predicates in their unmarked valence are normally realised as core arguments*
 - The core–oblique distinction has to do with the morphosyntactic encoding of dependents, not with their status as obligatory or selected by the predicate

What is UD?

- Challenges

- Handling ambiguous cases
- Language-specific phenomena
 - (de Marneffe et al., 2021) “*One of the main challenges for a framework like UD is to ensure that universal categories are applied consistently across languages with sometimes radically different morphosyntactic encoding strategies. This can only be achieved through a complex interplay between abstract language-independent guidelines and concrete language-specific criteria.*”

- Best practices

- Consistency, **clear documentation**, annotation cross-check

CoNLL-U format

```
117
118 # sent_id = weblog-blogger.com_marketview_20050511222700_ENG_20050511_222700-0003
119 # text = Google is a nice search engine.
120 1      Google Google  PROPN  NNP      Number=Sing      6      nsubj      6:nsubj _
121 2      is      be      AUX    VBZ      Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin  6      cop      6:cop _
122 3      a      a      DET    DT      Definite=Ind|PronType=Art      6      det      6:det _
123 4      nice    nice    ADJ    JJ      Degree=Pos      6      amod      6:amod _
124 5      search  search  NOUN   NN      Number=Sing      6      compound      6:compound _
125 6      engine  engine  NOUN   NN      Number=Sing      0      root      0:root SpaceAfter=No
126 7      .      .      PUNCT  .      _      6      punct      6:punct _
127
128 # sent_id = weblog-blogger.com_marketview_20050511222700_ENG_20050511_222700-0004
129 # text = Does anybody use it for anything else?
130 1      Does    do      AUX    VBZ      Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin  3      aux      3:aux _
131 2      anybody anybody PRON   NN      Number=Sing|PronType=Ind      3      nsubj      3:nsubj _
132 3      use      use      VERB   VB      VerbForm=Inf      0      root      0:root _
133 4      it      it      PRON   PRP      Case=Acc|Gender=Neut|Number=Sing|Person=3|PronType=Prs  3      obj      3:obj _
134 5      for      for      ADP    IN      _      6      case      6:case _
135 6      anything anything PRON   NN      Number=Sing|PronType=Ind      3      obl      3:obl:for _
136 7      else    else    ADJ    JJ      Degree=Pos      6      amod      6:amod SpaceAfter=No
137 8      ?      ?      PUNCT  .      _      3      punct      3:punct _
138
```

CoNLL-U format

- Sentences consist of one or more word lines, and word lines contain the annotation of a word/token/node in 10 fields separated by single tab characters
 - ID, FORM, LEMMA, UPOS, XPOS, FEATS, HEAD, DEPREL, DEPS, MISC
- The last line of each sentence is a blank line
- Comment lines (with #) occur at the beginning of sentences, before word lines

```
117 # sent_id = weblog-blogger.com_marketview_20050511222700_ENG_20050511_222700-0003
118 # text = Google is a nice search engine.
119
120 1 Google Google PRON NNP Number=Sing 6 nsubj 6:nsubj _
121 2 is be AUX VBZ Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin 6 cop 6:cop _
122 3 a a DET DT Definite=Ind|PronType=Art 6 det 6:det _
123 4 nice nice ADJ JJ Degree=Pos 6 amod 6:amod _
124 5 search search NOUN NN Number=Sing 6 compound 6:compound _
125 6 engine engine NOUN NN Number=Sing 0 root 0:root SpaceAfter=No
126 7 . . PUNCT . _ 6 punct 6:punct _
127
128 # sent_id = weblog-blogger.com_marketview_20050511222700_ENG_20050511_222700-0004
129 # text = Does anybody use it for anything else?
130
131 1 Does do AUX VBZ Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin 3 aux 3:aux _
132 2 anybody anybody PRON NN Number=Sing|PronType=Ind 3 nsubj 3:nsubj _
133 3 use use VERB VB VerbForm=Inf 0 root 0:root _
134 4 it it PRON PRP Case=Acc|Gender=Neut|Number=Sing|Person=3|PronType=Prs 3 obj 3:obj _
135 5 for for ADP IN _ 6 case 6:case _
136 6 anything anything PRON NN Number=Sing|PronType=Ind 3 obl 3:obl:for _
137 7 else else ADJ JJ Degree=Pos 6 amod 6:amod SpaceAfter=No
138 8 ? ? PUNCT . _ 3 punct 3:punct _
```

CoNLL-U format

- Annotation fields
 - ID: Word index, integer starting at 1 for each new sentence
 - FORM: Word form or punctuation symbol
 - LEMMA: Lemma or stem of word form

```
118 # sent_id = weblog-blogspot.com_marketview_20050511222700_ENG_20050511_222700-0003
119 # text = Google is a nice search engine.
120 1 Google Google PROPN NNP Number=Sing 6 nsubj 6:nsubj _
121 2 is be AUX VBZ Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin 6 cop 6:cop _
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124 5 search search NOUN NN Number=Sing 6 compound 6:compound _
125 6 engine engine NOUN NN Number=Sing 0 root 0:root SpaceAfter=No
126 7 . . PUNCT . _ 6 punct 6:punct _
```


CoNLL-U format

- Annotation fields
 - UPOS: Universal part-of-speech tag
 - XPOS: Language-specific (or treebank-specific) part-of-speech / morphological tag; underscore if not available
 - FEATS: List of morphological features from the universal feature inventory or from a defined language-specific extension; underscore if not available

```
118 # sent_id = weblog-blogspot.com_marketview_20050511222700_ENG_20050511_222700-0003
119 # text = Google is a nice search engine.
120 1      Google Google  PROPN  NNP      Number=Sing  6      nsubj  6:nsubj  _
121 2      is      be      AUX    VBZ      Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin  6      cop      6:cop  _
122 3      a      a      DET    DT        Definite=Ind|PrtnType=Art  6      det      6:det  _
123 4      nice    nice    ADJ    JJ        Degree=Pos  6      amod     6:amod  _
124 5      search  search  NOUN   NN        Number=Sing  6      compound 6:compound _
125 6      engine  engine  NOUN   NN        Number=Sing  0      root     0:root  SpaceAfter=No
126 7      .      .      PUNCT  .         _          6      punct   6:punct  _
```

CoNLL-U format

- Annotation fields

- HEAD: Head of the current word, which is either a value of ID or zero (0, root)
- DEPREL: Universal dependency relation to the HEAD or a defined language-specific subtype of one
- DEPS: Enhanced dependency graph in the form of a list of head-deprel pairs
- MISC: Any other annotation

```
118 # sent_id = weblog-blogspot.com_marketview_20050511222700_ENG_20050511_222700-0003
119 # text = Google is a nice search engine.
120 1      Google Google  PROPN  NNP      Number=Sing  6      nsubj  6:nsubj  _
121 2      is      be      AUX    VBZ      Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin  6      cop      6:cop  _
122 3      a      a      DET    DT        Definite=Ind|PronType=Art  6      det      6:det  _
123 4      nice    nice    ADJ    JJ        Degree=Pos  6      amod     6:amod  _
124 5      search  search  NOUN   NN        Number=Sing  6      compound 6:compound  _
125 6      engine  engine  NOUN   NN        Number=Sing  0      root     0:root  SpaceAfter=No
126 7      .      .      PUNCT  .         _         6      punct   6:punct  _
```

CoNLL-U format

- Non-English datasets

```
525 # comment = node 2 mistagged in ARCOSG
526 # sent_id = c01_034b
527 # speaker = [3]
528 # text = 's e an taobh ceàrr a bh' ann
529 1      's      is      AUX      Wp-i      Tense=Pres      4      cop      _      _
530 2      e      e      PRON      Pp3sm      Gender=Masc|Number=Sing|Person=3      1      fixed      _      _
531 3      an      an      DET      Tdsm      Definite=Def|Gender=Masc|Number=Sing|PronType=Art      4      det      _      _
532 4      taobh    taobh    NOUN      Ncsmn      Case=Nom|Gender=Masc|Number=Sing      0      root      _      _
533 5      ceàrr    ceàrr    ADJ      Aq-smn      Case=Nom|Gender=Masc|Number=Sing      4      amod      _      _
534 6      a      a      PART      Q-r      PartType=Vb|PronType=Rel      7      nsubj      _      _
535 7      bh'      bi      VERB      V-s      Tense=Past      4      csubj:cleft      _      _
536 8-9     ann      _      _      _      _      _      _      _      _
537 8      an      an      ADP      Sp      _      9      case      _      _
538 9      e      e      PRON      Pp3sm      Gender=Masc|Number=Sing|Person=3      7      xcomp:pred      _      _
```

CoNLL-U format

- Non-English datasets

```
# sent_id = n01001013
# text = สำหรับผู้ที่ติดตามการเปลี่ยนผ่านโซเชียลมีเดียในแคปิตอลฮิลล์ เรื่องนี้จะแตกต่างกันเล็กน้อย
# translit = sǎhrâbpʰú²dʰí¹tidtámkárpélí¹ynpʰá¹ncóćʰíálmídíánaigʰápitoːlhil rúá¹ŋní²čatáktá¹ŋpailekn²'y
# text_en = For those who follow social media transitions on Capitol Hill, this will be a little different.
1      สำหรับ      _      ADP      IN      _      2      case      _      SpaceAfter=No|Translit=sǎhrâb
2      ผู้      _      NOUN     NN      _      13     obl      _      SpaceAfter=No|Translit=pʰú²
3      ที่      _      DET      WDT     _      4      nsubj   _      SpaceAfter=No|Translit=dʰí¹
4      ติดตาม      _      VERB     VV      _      2      acl:relcl _      SpaceAfter=No|Translit=tidtám
5      การเปลี่ยนผ่าน      _      VERB     VV      _      4      obj      _      SpaceAfter=No|Translit=kárpélí¹ynpʰá¹n
6      โซเชียล      _      ADJ      JJ      _      7      amod     _      Proper=True|SpaceAfter=No|Translit=cóćʰíál
7      มีเดีย      _      NOUN     NN      _      5      obj      _      SpaceAfter=No|Translit=mídíá
8      ใน      _      ADP      IN      _      9      case     _      SpaceAfter=No|Translit=nai
9      แคปิตอลฮิลล์      _      PROPN    NNP     _      5      obl      _      Translit=gʰápitoːlhil
10     เรื่อง      _      NOUN     NN      _      13     nsubj   _      SpaceAfter=No|Translit=rúá¹ŋ
11     นี่      _      DET      DT      _      10     det      _      SpaceAfter=No|Translit=ní²
12     จะ      _      AUX      MD      _      13     aux      _      SpaceAfter=No|Translit=ča
13     ต่าง      _      VERB     VV      _      0      root     _      SpaceAfter=No|Translit=táktá¹ŋ
14     ไป      _      PART     RP      _      13     compound:prt _      SpaceAfter=No|Translit=pai
15     เล็กน้อย      _      ADV      RB      _      13     advmod   _      Translit=lekn²'y
```

CoNLL-U format

- Non-English datasets

```
10      # sent_id = 2
11      # text = 咪執返啲嘢去阿哥個新屋度嚟。
12      1      咪      咪      ADV      _      _      2      advmod      _      SpaceAfter=No
13      2      執      執      VERB     _      _      0      root      _      SpaceAfter=No
14      3      返      返      VERB     _      _      2      compound:dir _      SpaceAfter=No
15      4      啲      啲      NOUN     _      NounType=Clf 5      clf:det      _      SpaceAfter=No
16      5      嘢      嘢      NOUN     _      _      3      obj      _      SpaceAfter=No
17      6      去      去      VERB     _      _      2      conj      _      SpaceAfter=No
18      7      阿哥    阿哥    NOUN     _      _      10     nmod      _      SpaceAfter=No
19      8      個      個      NOUN     _      NounType=Clf 10     clf:det      _      SpaceAfter=No
20      9      新      新      ADJ      _      _      10     amod      _      SpaceAfter=No
21      10     屋      屋      NOUN     _      _      6      obj      _      SpaceAfter=No
22      11     度      度      ADP      _      _      10     case:loc    _      SpaceAfter=No
23      12     嚟      嚟      PART     _      _      2      discourse:sp _      SpaceAfter=No
24      13     。      。      PUNCT    _      _      2      punct      _      SpaceAfter=No
```

Mapping Korean Morphosyntax to UD

- <https://universaldependencies.org/guidelines.html>
- <https://nlpxl2korean.github.io/UD-KSL/xpos/>
- <https://nlpxl2korean.github.io/UD-KSL/ud/>
- https://github.com/NLPxL2Korean/UD-KSL/blob/main/dataset/xpos_upos_mapping_db_v1.0.txt

Available Korean UD Treebanks

parsed & annotated corpus

- Check the UD website!

Available Korean UD Treebanks

parsed & annotated corpus

	Korean	5	615K	🎓📅📝📖📐📏	Korean
Korean treebanks					
▶	KSL	155K	Ⓛ	📝	CC BY SA ★★★★★
▶	Kaist	350K	Ⓛ	🎓📖📐	CC BY SA ★★★★★
▶	GSD	80K	Ⓛ	📅📖	CC BY SA ★★★★★
▶	PUD	16K	Ⓛ ⓕ Ⓟ	📖📐	CC BY SA ★★★★★
▶	LittlePrince	13K	Ⓛ ⓕ Ⓟ	📖	CC BY SA ★★★★★

See [here](#) for comparative statistics of Korean treebanks.

Available Korean UD Treebanks

parsed & annotated corpus

 **KSL** **155K**    ★★★★★

UD_Korean-KSL is a dependency treebank of second-language (L2) Korean.

- Contributors: Hakyung Sung, Gyu-Ho Shin
- Repository [master](#) [dev](#)
- [README](#)
- [Treebank hub page](#)
- [Download](#)

Available Korean UD Treebanks

parsed & annotated corpus

- Check the UD website!
- Why UD fits workshop goals:
 - **Unified framework for L1 & L2:** applies identical structural schemas to native and learner writing
 - **Cross-lingual comparison:** enables direct comparison across languages
 - **Queries beyond frequency:** facilitates extracting complex morphosyntactic configurations (e.g., non-canonical word order, particle drop, light verbs) rather than simple word counts

Interactive Hands-On Exploration & Group Discussion

Hands-on exploration: UDPipe online parser

- <https://lindat.mff.cuni.cz/services/udpipe/>

- Step-by-step instructions:

- Input sentence
- Model 1 selection: korean-gsd-ud-2.12
- Output
- Model 2 selection: korean-ksl-ud-2.12
- Output & comparison

- 그래서 친구하고 병원에 갔습니다.
- 외국어를 학교이나 테키스트으로 공부하는 아니고 자연히 말 할 수 있다.
- 그 사람과 잘 어울리면 일을 더 빠르게 하고 아이디어가 많고 좋은 결과가 나온다.

Group discussion

- How to annotate?
 - Topic marker attachment in multiple subject constructions
 - Does *-un/-nun* always mark **nsubj**?
 - Sentence A: 나는 사과를 먹었다.
 - Sentence B: 사과는 내가 먹었다.

Group discussion

- How to annotate?
 - Light verb constructions
 - *How does token spacing alter the **X/UPOS tag** and **dependency tag**?*
 - Sentence A: 나는 어제 공부했다.
 - Sentence B: 나는 어제 공부를 했다.

Group discussion

- How to annotate?
 - Copula verb
 - *How is the attached copula -i- annotated?*
 - Sentence: 인호는 학생이다.
 - <https://universaldependencies.org/u/dep/cop.html>
 - <https://universaldependencies.org/u/dep/root.html>